AnyCompany Bank

Retail Payments Platform: System Design Document

Document ID: SDD-ACB-2026-005 · Version 1.3 · Service tier: Tier 1 · Hosting: on-premises (owned datacenters) · Classification: Internal (sample data)

1 Overview

The Retail Payments Platform authorises and settles card and account-to-account payments for AnyCompany Bank retail customers. It is classified Tier 1 under ACB-STD-04: an outage stops customer payments. The platform runs entirely in the bank's own datacenters: Site A in Frankfurt (DC-FRA) and Site B in Amsterdam (DC-AMS), 360 km apart on separate grids and carrier entry points. A cold standby site in Dublin (DC-DUB) holds backups.

2 Site topology

Figure 1 shows the estate in the ACB-STD-04 house notation; see the notation sheet for symbol meanings. Both sites run core banking application nodes behind a traffic director pair and serve traffic at all times. Each site has a core database, a SAN array and a hardware security module for payment signing. All inbound traffic enters through the internal API gateway.

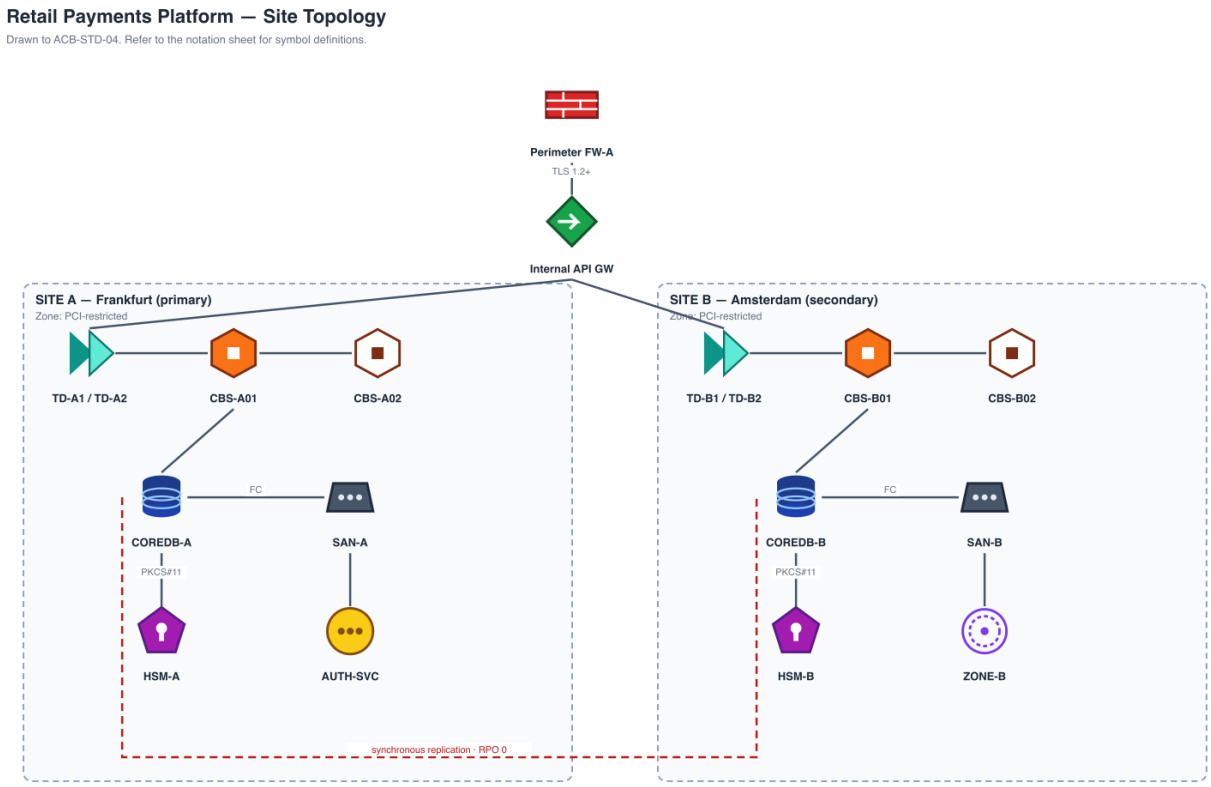


Figure 1: Retail Payments Platform site topology (ACB-STD-04 notation).

3 Data and replication

COREDB-A and COREDB-B replicate synchronously, giving a recovery point objective of zero. Each database is backed by a fibre-channel SAN array at its own site. Nightly backups are copied to the cold site in Dublin.

4 Disaster recovery

Figure 2 shows the failover path. On loss of Site A, the Global Traffic Director routes all traffic to Site B and COREDB-B is promoted. The most recent full site failover test was completed on 14 January 2026, with a measured recovery time of 22 minutes. Site failover tests are run quarterly.

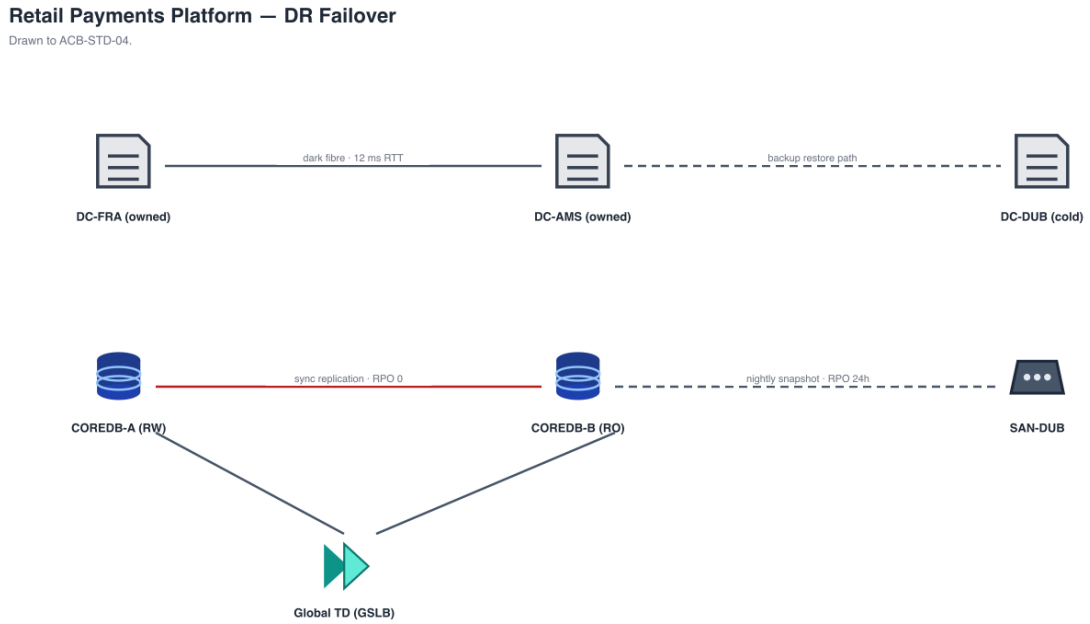


Figure 2: Site-level failover and replication paths (ACB-STD-04 notation).

5 Shared services

The platform consumes the shared authentication service (AUTH-SVC) and the shared notification service. The notification service is currently classified Tier 2.

6 Known risks

Risk	Status
Site failover is tested quarterly; the last measured recovery time was 22 minutes.	Open
The shared notification service is Tier 2, below its Tier 1 consumer.	Open